

CsI(Tl) scintillation crystals

Description: Caesium Iodide doped with Thallium (CsI(Tl)) Scintillation crystal

Scintillation crystal	:	CsI(Tl)
Physical form	:	Single crystal
Crystal class	:	Cubic, No cleavage
Chemical nature	:	CsI(Tl)

Physical properties

Density	:	4.51 g/cm ³
Max emission wavelength	:	550 nm
Decay time	:	600 ns, 3.4 μ s
Refractive index at emission max.	:	1.79
Photoelectron yield with bi-alkali	:	45% relative to NaI(Tl)
Optical attenuation coefficient	:	> 1 m
Thallium concentration	:	0.1 – 0.12 mole%
Useful temperature range	:	-40°C to +125°C
Storage conditions	:	< 50% relative humidity when bare
Thermal conductivity	:	0.13 W.m ⁻¹ .K ⁻¹
Thermal expansion	:	51.10 ⁻⁶ K ⁻¹
Specific heat	:	201 J/kg.K
Apparent Elastic limit	:	5.58 MPa
Young's modulus	:	5.3 GPa
Shear modulus	:	6.24 GPa
Bulk modulus	:	12.7 GPa
Maximum thermal gradient	:	Depending on the size of a crystal For small cm size crystals: 10 degrees/min

Peculiarities of SCIONIX CsI(Tl) crystals

SCIONIX CsI(Tl) crystals are grown using a specific process that guarantees a very homogeneous Thallium concentration over the entire crystal volume. This allows an extremely good energy resolution and homogeneity of response. All crystals are selected on light output.

and resolution. The crystal wrapping is certified to maximize the light output. CsI(Tl) crystals are used in many calorimeters like MUST, Chimers, HIRA etc.

Our CsI(Tl) crystals are unique in constant factor of the doping concentration and the fact that all crystals are selected on this.

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